



Generalized Rank-based Evaluation for Knowledge Graph Completion: Perspectives, Framework, and Analyses

2026-07-22

MINDS Lab

SooHo Moon



Chung-Ang University, South Korea

Jian Kang

Mohamed bin Zayed University
of Artificial Intelligence, UAE

YunYong Ko*

Chung-Ang University, South Korea

INDEX

■ Knowledge Graph

■ KGC Evaluation & Motivation

■ PROBE Framework

■ Experiments

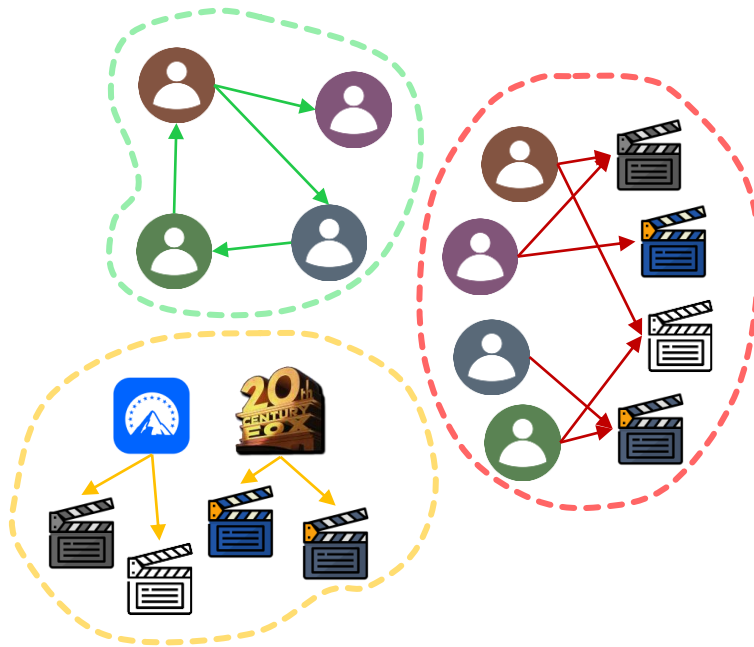
■ Conclusion

KNOWLEDGE GRAPH

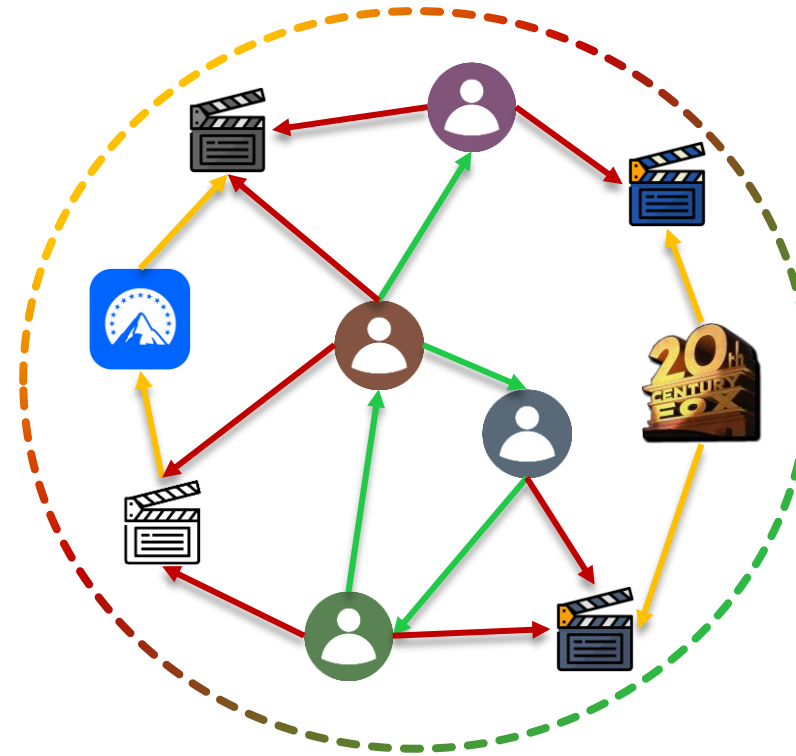
■ What is a Knowledge Graph (KG)?

□ Edge : **Triple** (head entity, relation, tail entity)

[Ordinary Graph]



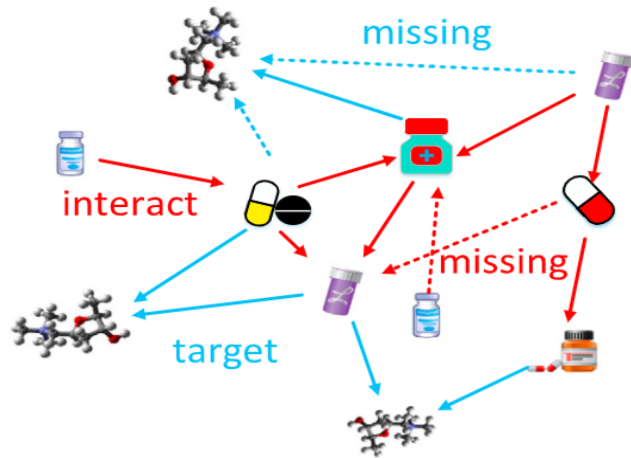
[Knowledge Graph]



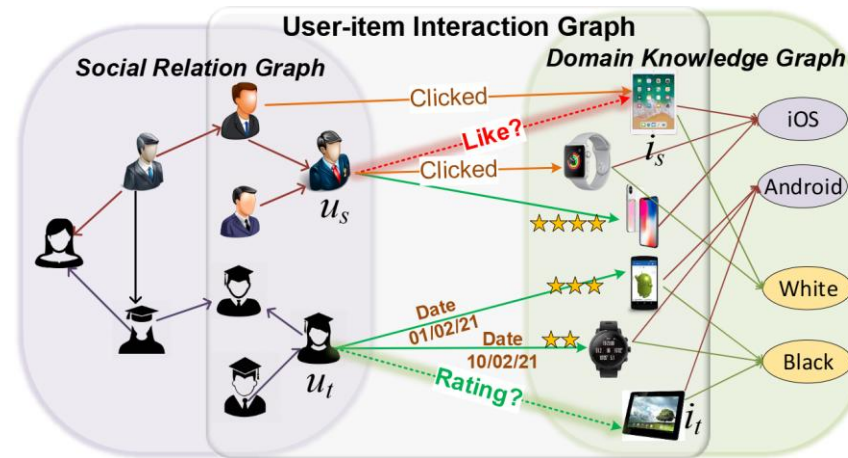
KNOWLEDGE GRAPH

■ Used in Various Real-life Applications

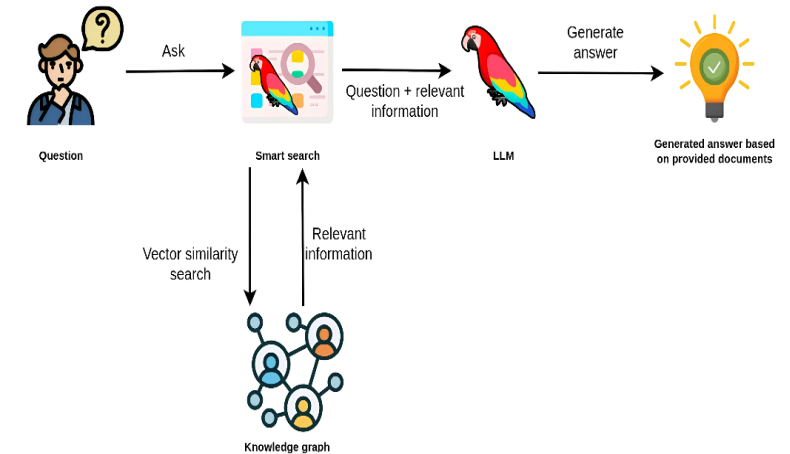
<Drug Discovery>



<Recommender System>



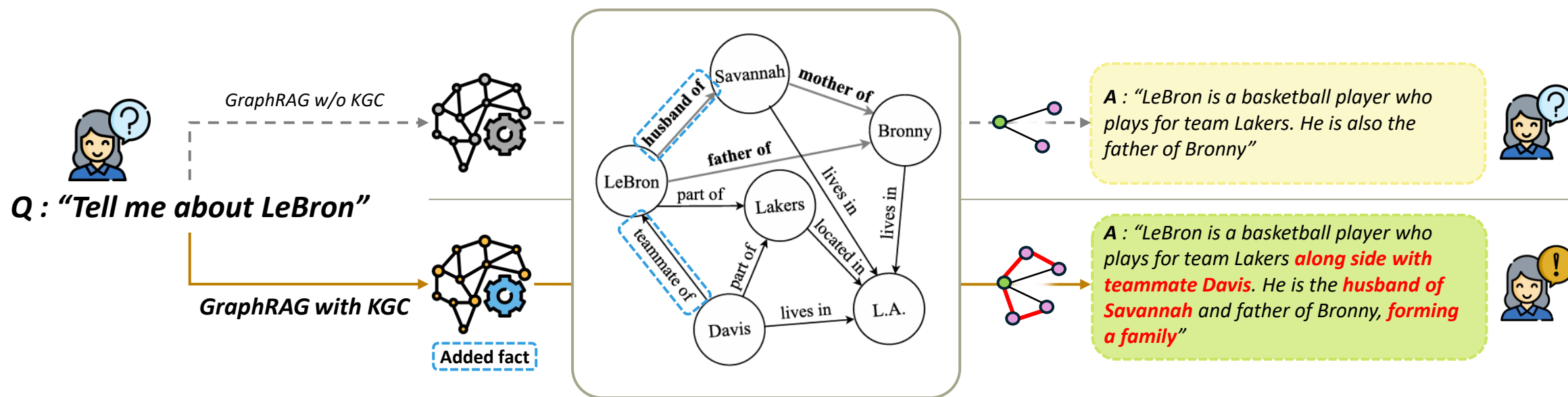
<GraphRAG>



KNOWLEDGE GRAPH

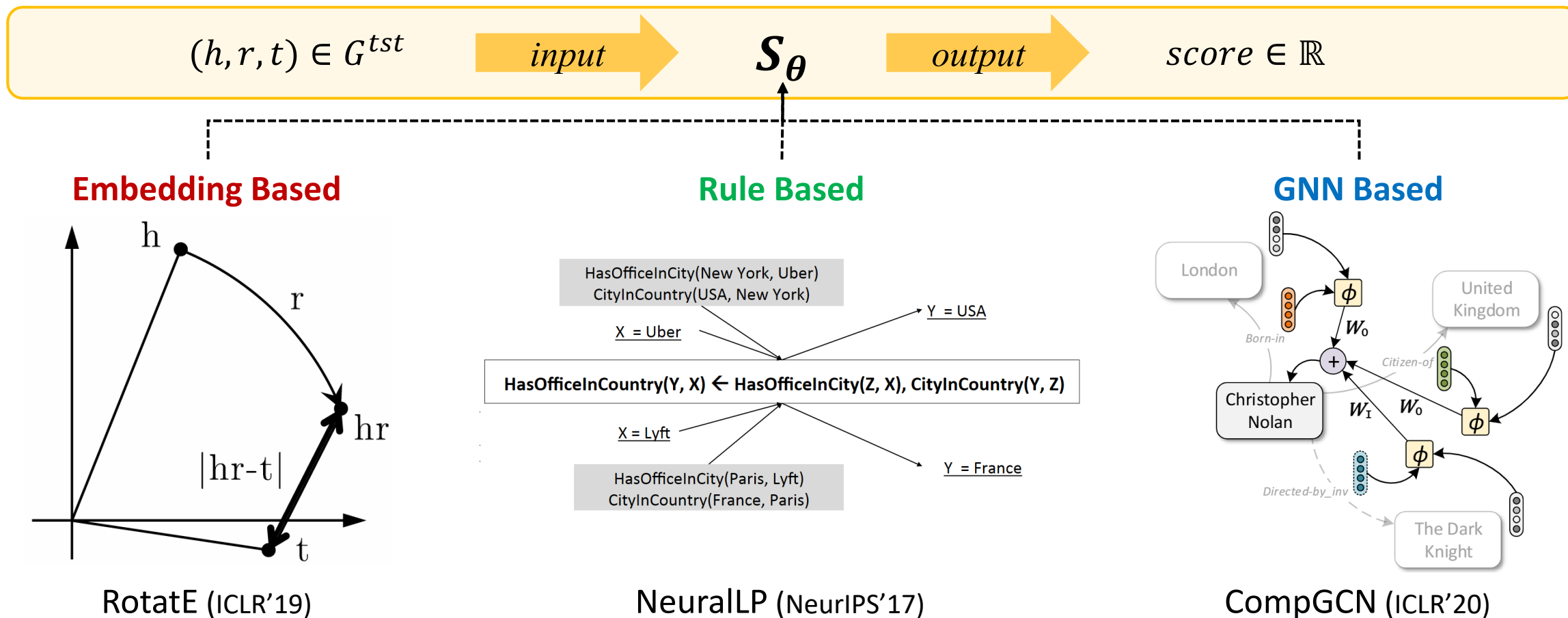
■ Knowledge Graph Completion (KGC)

- Highly sparse KG → greatly hinder performance of systems that rely upon it
- **Knowledge Graph Completion** : $(h, r, ?)$ or $(?, r, t) \rightarrow \text{predict “?”}$



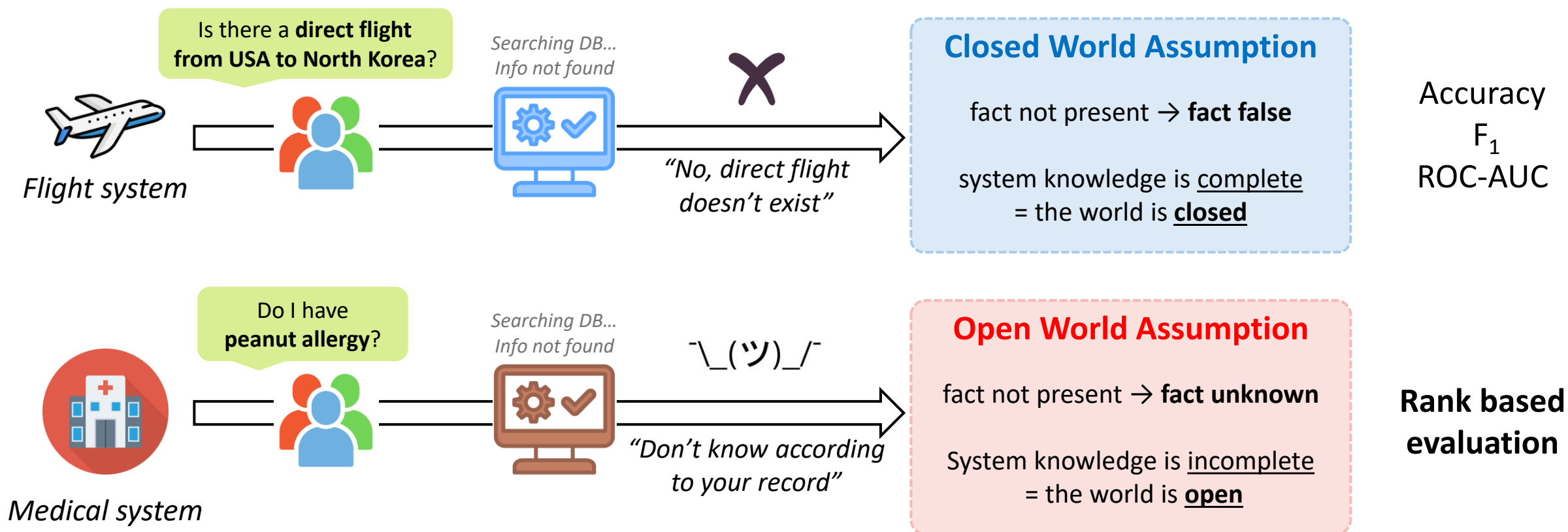
KNOWLEDGE GRAPH

■ Various KGC Methods



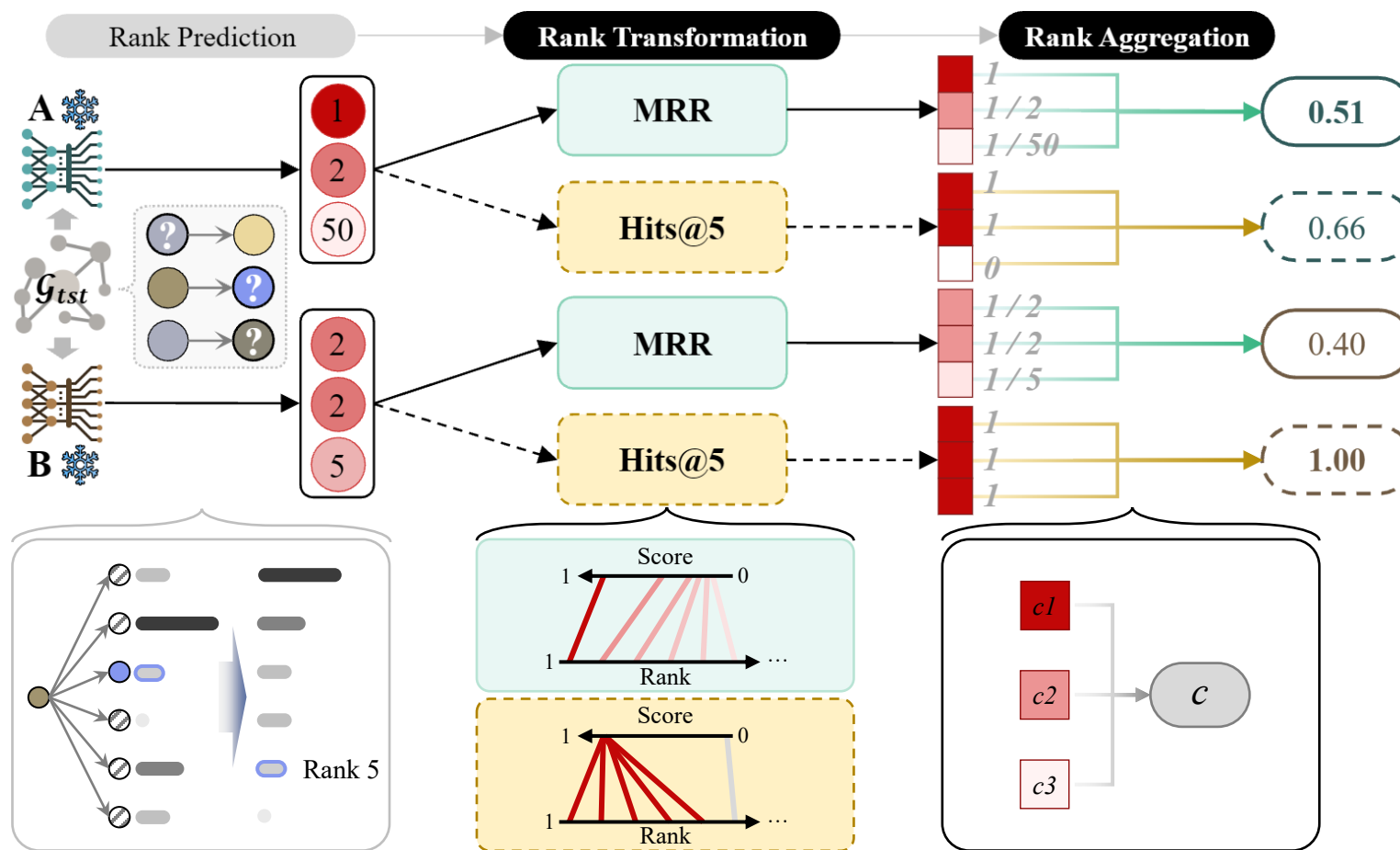
KGC EVALUATION & MOTIVATION

■ What Metrics Should We Use?



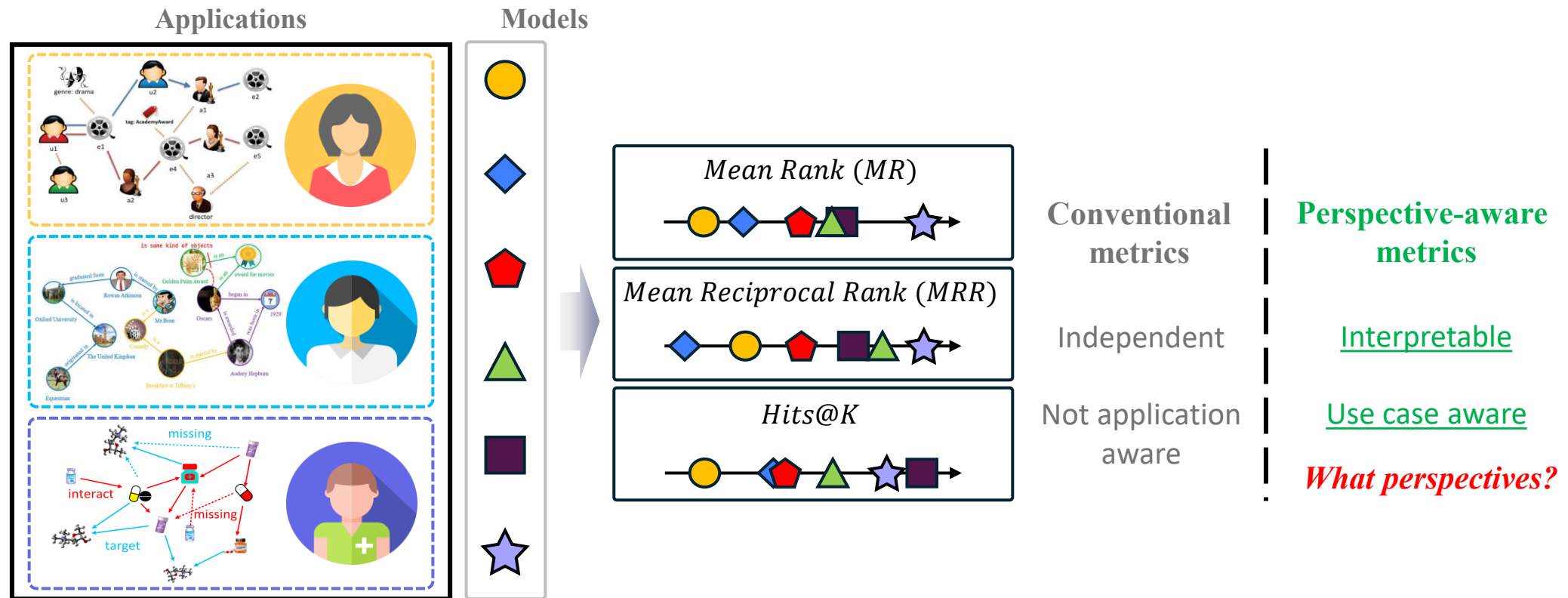
KGC EVALUATION & MOTIVATION

Rank Based Metrics



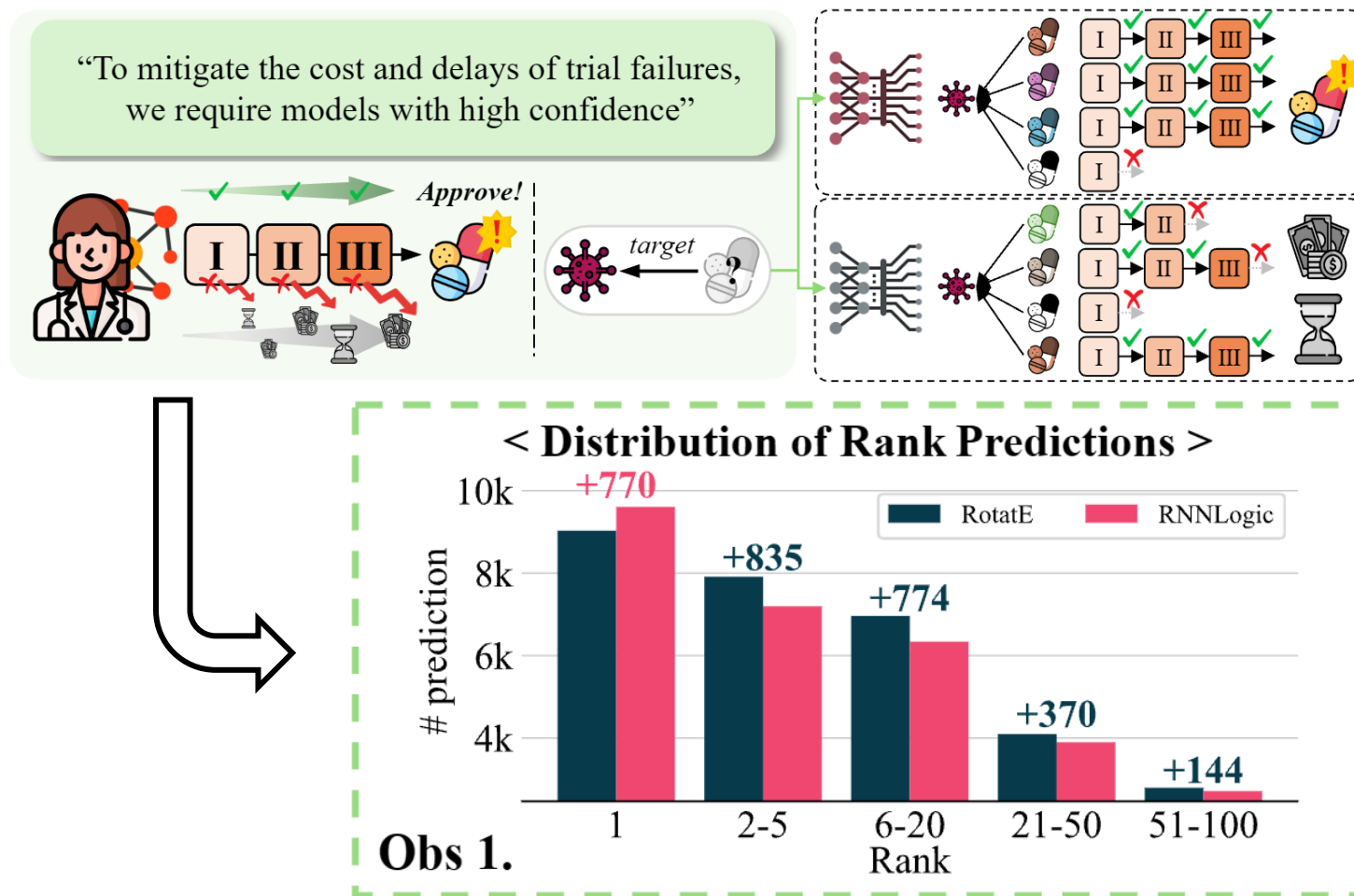
KGC EVALUATION & MOTIVATION

Shortcomings of Conventional Metrics



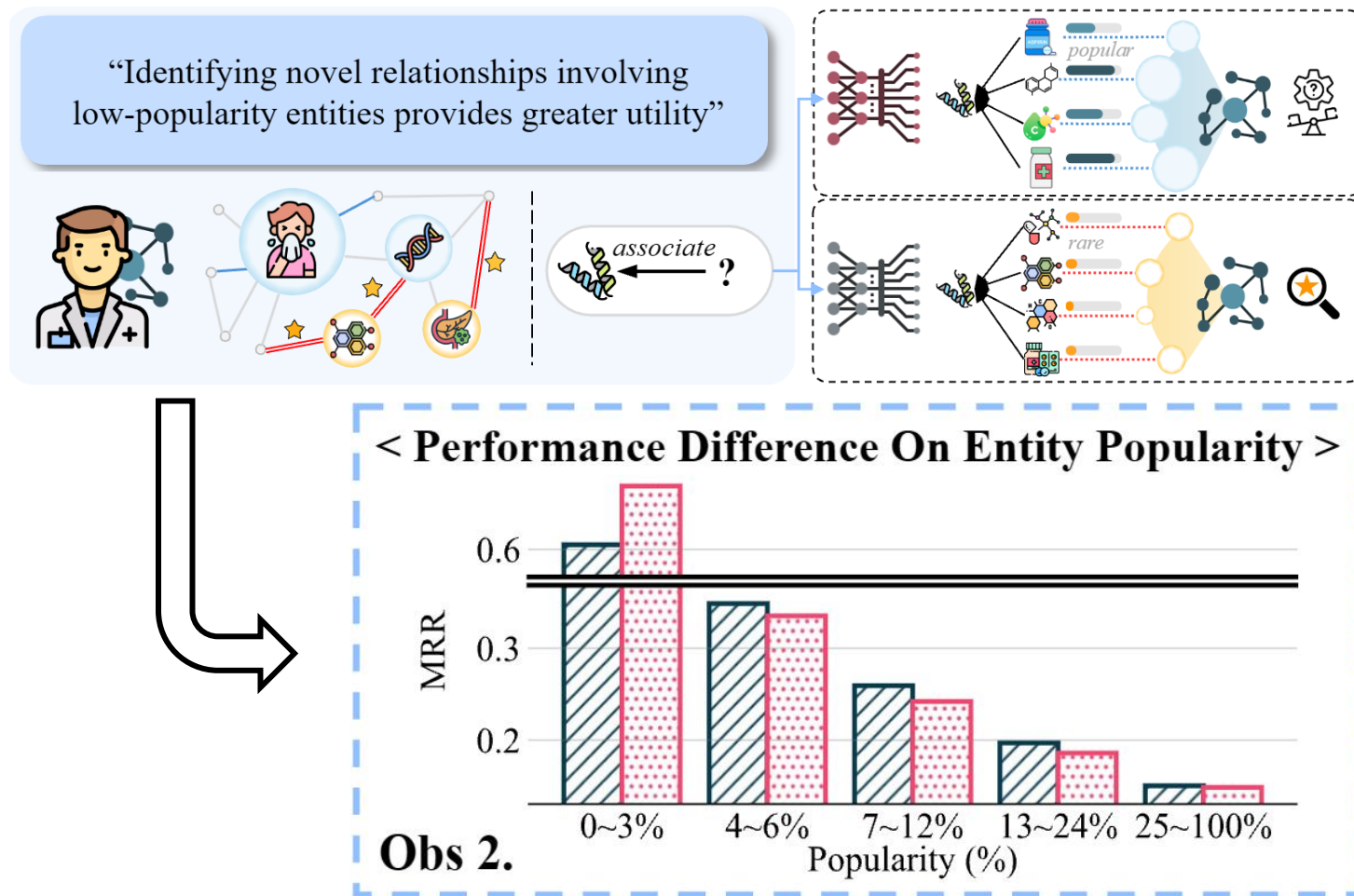
KGC EVALUATION & MOTIVATION

■ (P1) Predictive Sharpness



KGC EVALUATION & MOTIVATION

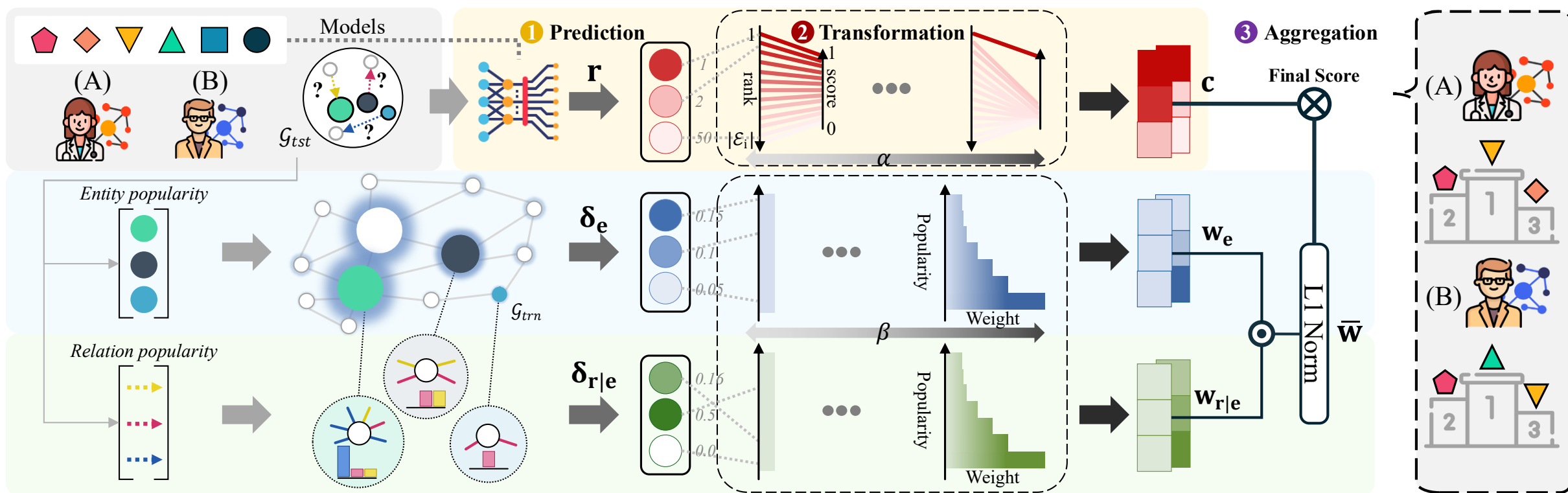
■ (P2) Popularity-bias Robustness



PROBE FRAMEWORK

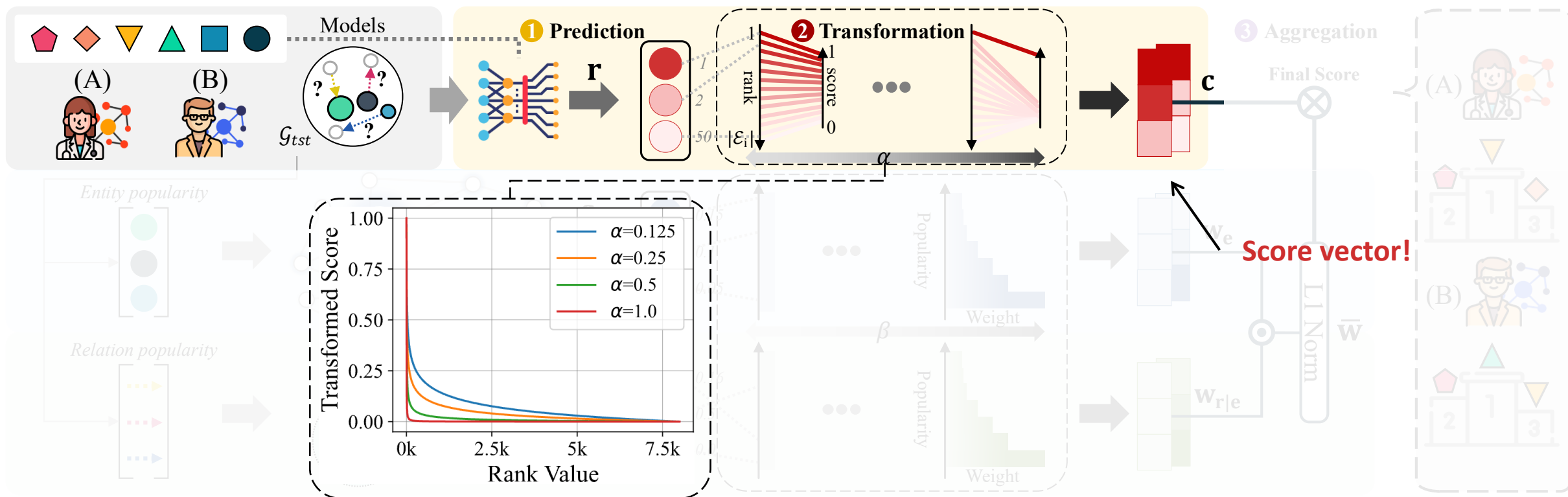
Overview of PROBE Framework

Predictive shaRpness and pOpopularity-Bias robustness aware Evaluation



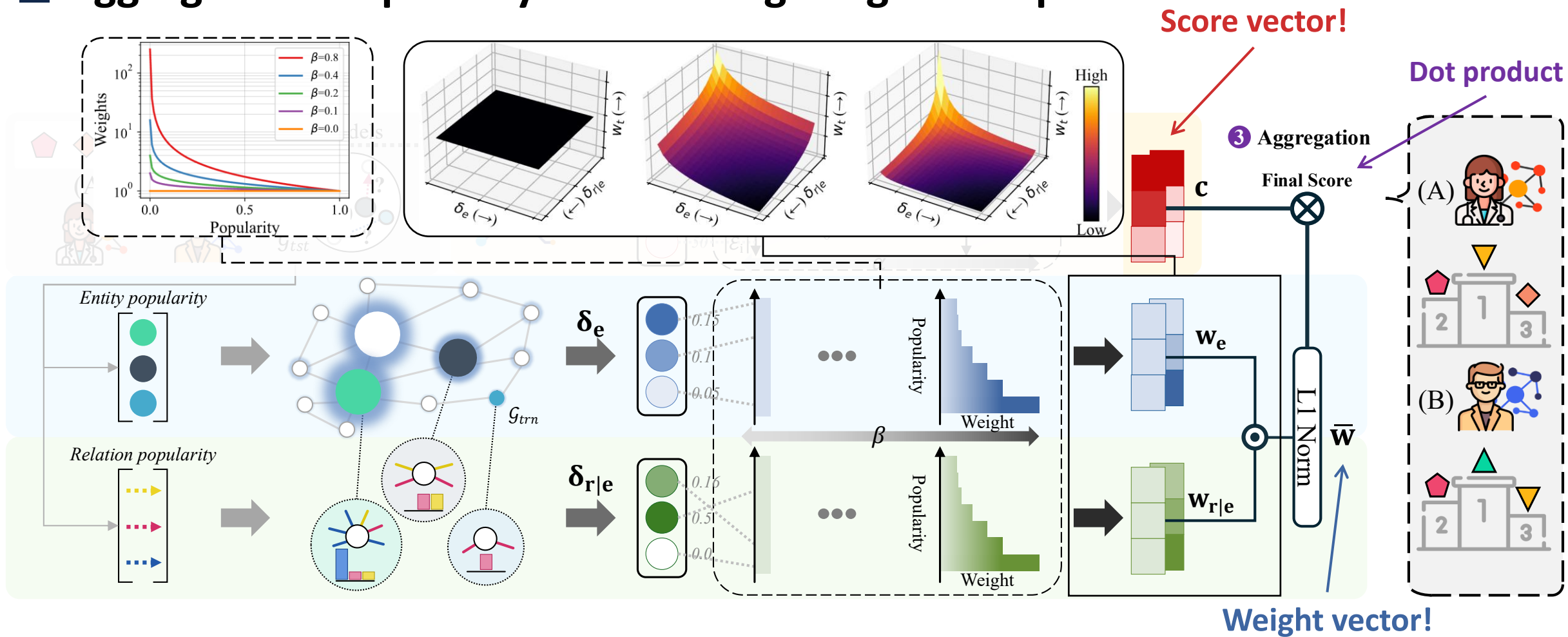
PROBE FRAMEWORK

Transformation: Predictive Sharpness Control Factor α



PROBE FRAMEWORK

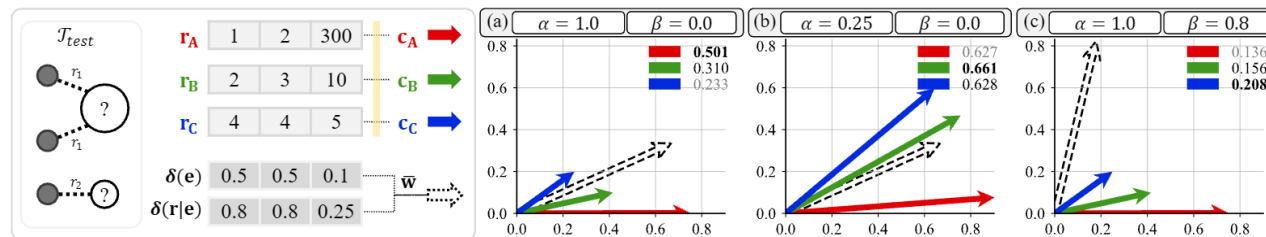
■ Aggregation: Popularity-aware Weighting Factor β



PROBE FRAMEWORK

Comprehensive Analysis on PROBE

“How does PROBE produce different evaluations on varying perspectives?”



“Is PROBE a desirable rank-based metric?”

Property	MR	MRR	Hits@K	p-MRR	Strat-MRR	$\overline{WMR}$	AMRI	PROBE
FIXED OPTIMUM	✓	✓	✓	✓	✓	✓	✓	✓
FIXED PESSIMUM	-	-	✓	-	-	✓	✓	✓
ANTI MONOTONICITY	-	✓	-	✓	✓	✓	✓	✓
CANDIDATE-SIZE AWARENESS	-	-	-	-	-	✓	✓	✓
SHARPNESS AWARENESS	-	-	✓	✓	-	-	-	✓
POPULARITY AWARENESS	-	-	-	-	-	-	-	✓

“Does PROBE produce reliable evaluation under incomplete KGs?”

LEMMA 1. Let ℓ and α denote the true model strength under the CWA and the sharpness parameter, respectively. Then, the derivative of the expected PROBE score with respect to ℓ is a monotonically decreasing function of $\alpha \in \mathbb{R}$ ($\alpha \neq 0$):

$$\frac{\partial}{\partial \alpha} \left(\frac{d\hat{\mathbb{E}}(\text{PROBE})}{d\ell} \right) \leq 0.$$

■ Models, Datasets & Implementation Details

● RotatE

■ ComplEx

▲ HousE

▼ TuckER

◆ pLogicNet

◆ RNNLogic

○ Low dimension ($d=128$) tuned

○ Filtered setting, tie breaking protocol adopted for fair evaluation

○ Results are average of 3

$\alpha = [1.0, 0.5, 0.0, -0.5, -1.0]$

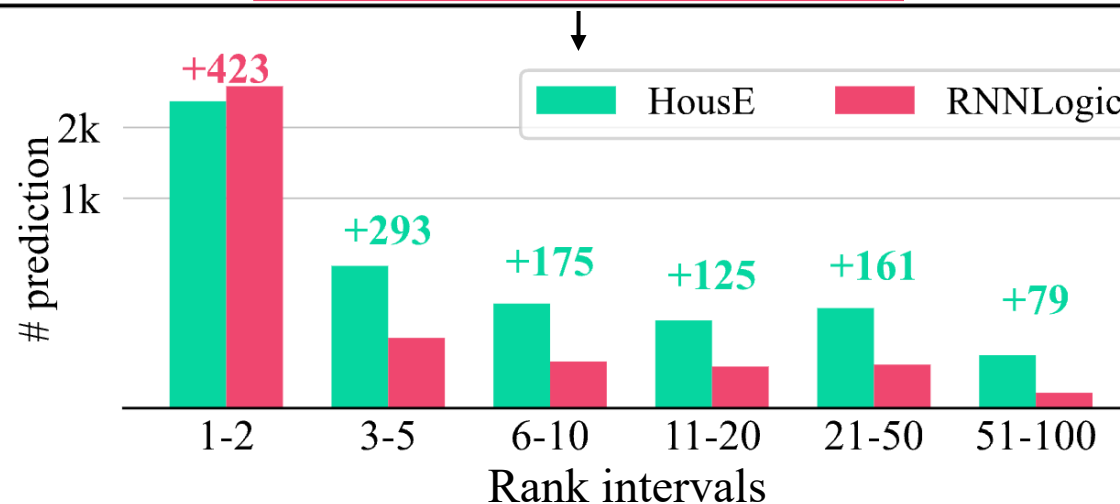
$\beta = [0.0, 0.2, 0.4, 0.6, 0.8]$

Dataset	$ \mathcal{E} $	$ \mathcal{R} $	$ \mathcal{G}_{trn} $	$ \mathcal{G}_{val} $	$ \mathcal{G}_{tst} $
FB15k237	14,541	237	272,115	17,535	20,466
WN18RR	40,943	11	86,835	3,034	3,134
YAGO3-10	123,182	37	1,079,040	5,000	5,000
Family	3,007	12	23,483	2,038	2,835
UMLS	135	46	1,959	1,306	3,264
Kinship	104	25	3,206	2,137	5,343

EXPERIMENTS

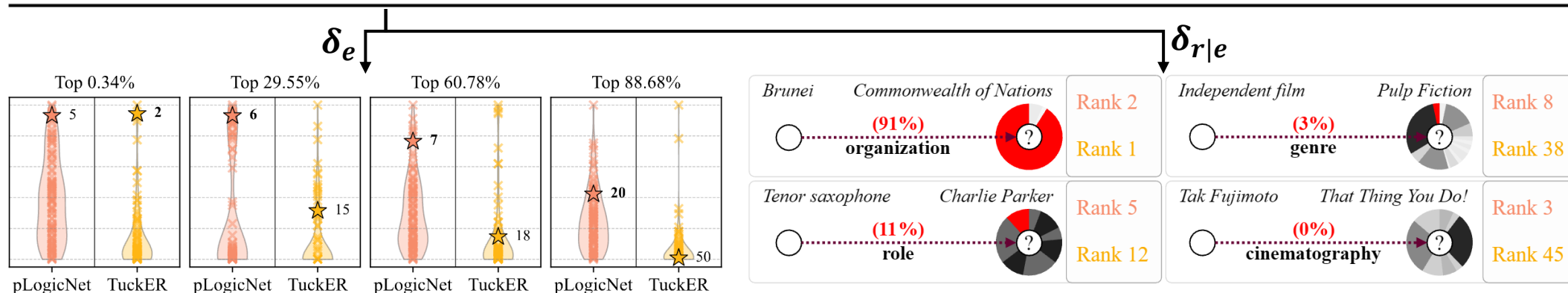
■ Varying Predictive Sharpness

Models	FB15k-237					WN18RR					YAGO3-10				
	$\alpha=1.0$	$\alpha=0.5$	$\alpha=0.0$	$\alpha=-0.5$	$\alpha=-1.0$	$\alpha=1.0$	$\alpha=0.5$	$\alpha=0.0$	$\alpha=-0.5$	$\alpha=-1.0$	$\alpha=1.0$	$\alpha=0.5$	$\alpha=0.0$	$\alpha=-0.5$	$\alpha=-1.0$
RotatE	0.3140	0.4241	0.7124	0.9380	0.9844	0.4641	0.5213	0.6952	0.8549	0.9017	0.4743	0.5640	0.7971	0.9555	0.9813
ComplEx	0.3146	0.4283	0.7213	0.9425	0.9855	0.4754	0.5179	0.6594	0.8121	0.8700	0.4496	0.5467	0.7979	0.9588	0.9815
HousE	0.3073	0.4100	0.6910	0.9244	0.9778	0.4184	0.4913	0.6991	0.8823	0.9346	0.4538	0.5297	0.7500	0.9267	0.9647
Tucker	0.3530	0.4608	0.7379	0.9479	0.9884	0.4625	0.4999	0.6368	0.7961	0.8583	0.5562	0.6246	0.8144	0.9550	0.9806
pLogicNet	0.3241	0.4311	0.7149	0.9386	0.9847	0.4957	0.5672	0.7593	0.9002	0.9292	0.4535	0.5467	0.7927	0.9576	0.9834
RNNLogic	0.3197	0.4218	0.6969	0.9213	0.9723	0.4807	0.5225	0.6567	0.7929	0.8401	OOM	OOM	OOM	OOM	OOM

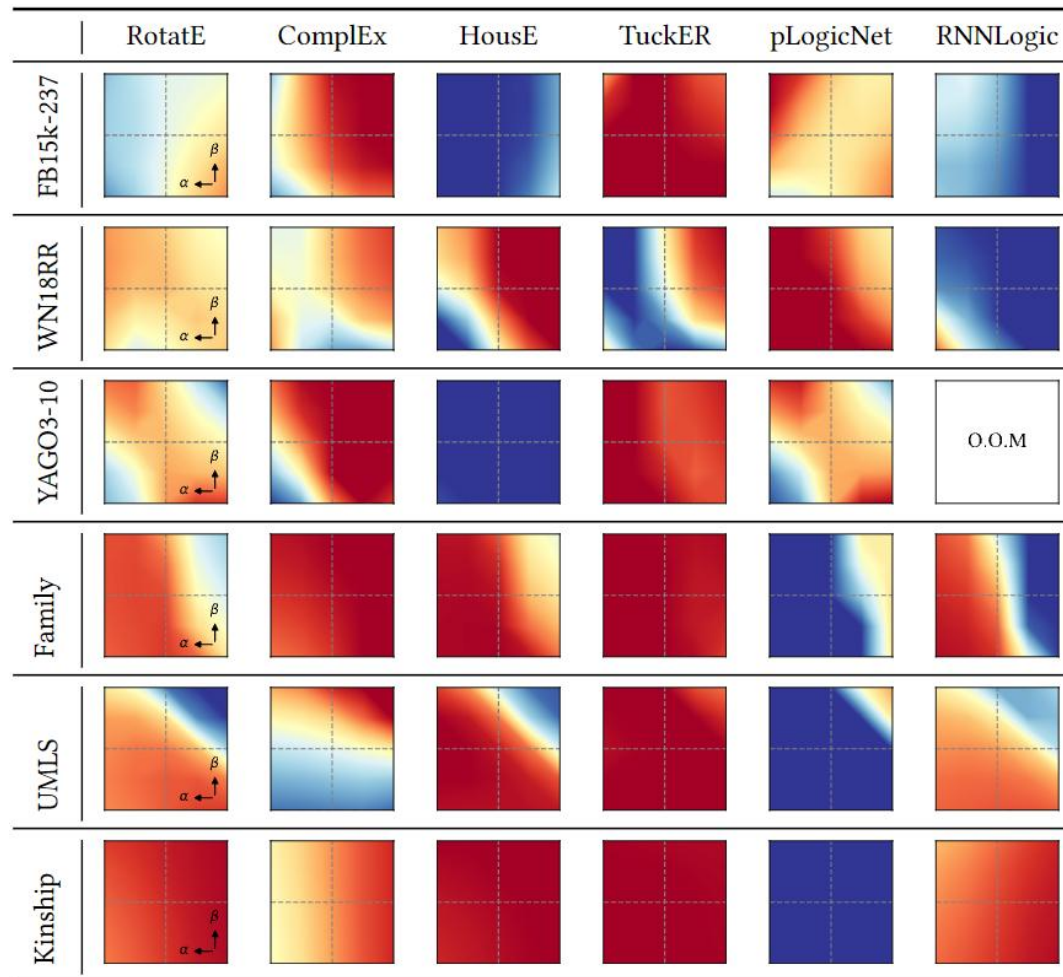


■ Varying Popularity-bias Robustness

Models	FB15k-237					WN18RR					YAGO3-10				
	$\beta=0.0$	$\beta=0.2$	$\beta=0.4$	$\beta=0.6$	$\beta=0.8$	$\beta=0.0$	$\beta=0.2$	$\beta=0.4$	$\beta=0.6$	$\beta=0.8$	$\beta=0.0$	$\beta=0.2$	$\beta=0.4$	$\beta=0.6$	$\beta=0.8$
RotatE	0.3140	0.1747	0.1043	0.0866	0.0820	0.4641	0.3326	0.2097	0.1462	0.1203	0.4743	0.3525	0.2207	0.1249	0.0791
ComplEx	0.3146	0.1801	0.1097	0.0897	0.0821	0.4754	0.3330	0.1998	0.1309	0.1029	0.4496	0.3476	0.2188	0.1241	0.0789
HousE	0.3073	0.1651	0.0948	0.0775	0.0731	0.4184	0.3017	0.1941	0.1383	0.1153	0.4538	0.3268	0.1828	0.0778	0.0297
TuckER	0.3530	0.2033	0.1286	0.1073	0.0980	0.4625	0.3125	0.1751	0.1058	0.0790	0.5654	0.4260	0.2675	0.1505	0.0947
pLogicNet	0.3241	0.1912	0.1244	0.1103	0.1097	0.4957	0.3478	0.2221	0.1601	0.1354	0.4535	0.3391	0.2167	0.1279	0.0861
RNNLogic	0.3197	0.1745	0.1058	0.0901	0.0862	0.4807	0.3243	0.1820	0.1099	0.0817	OOM	OOM	OOM	OOM	OOM



■ Dual Perspective Normalized Heatmap



Q&A